

Project Title Seasonal Agriculture - Performance Analysis

Student Name: **Manoj**

College Name: Seshadripuram Degree College

AICTE STU ID: STU6875e4756f7b3175255666



1

Data Visualization Project

PROBLEM STATEMENT



Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.

However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.

The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

PROJECT DESCRIPTION

Seasonal Agriculture Performance Analysis is a data-visualization project that compares Kharif, Rabi and Zaid agriculture.

The project uses a structured CSV dataset containing crop, season, area, yield, production, temperature and rainfall.

Python and Pandas are used for data processing, while Matplotlib is used to create visualizations.

The analysis helps identify production patterns, average yield and relationships between weather factors and crop performance.

WHO ARE THE END USERS?



Farmers - understand which seasons and crops perform better.

Agricultural students and researchers - study seasonal trends.

Farm planners - compare production and yield indicators.

Government/agriculture teams - use the dashboard concept for data-driven planning.

Data analysts - practice cleaning, grouping and visualization of agriculture data.

TECHNOLOGY USED

Python –main programming language.

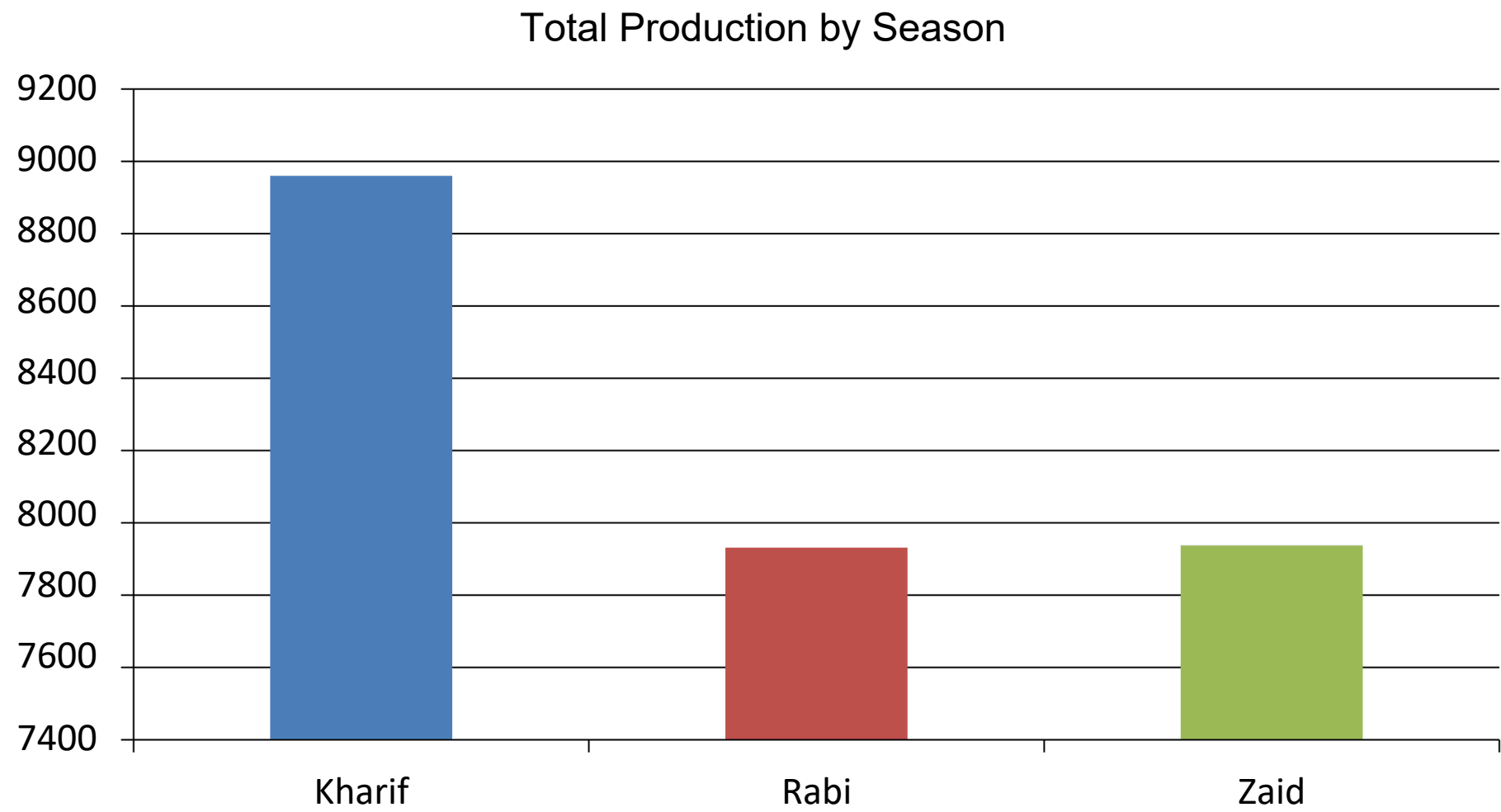
Pandas –data loading, cleaning, grouping and aggregation.

Matplotlib –bar charts and scatter plots.

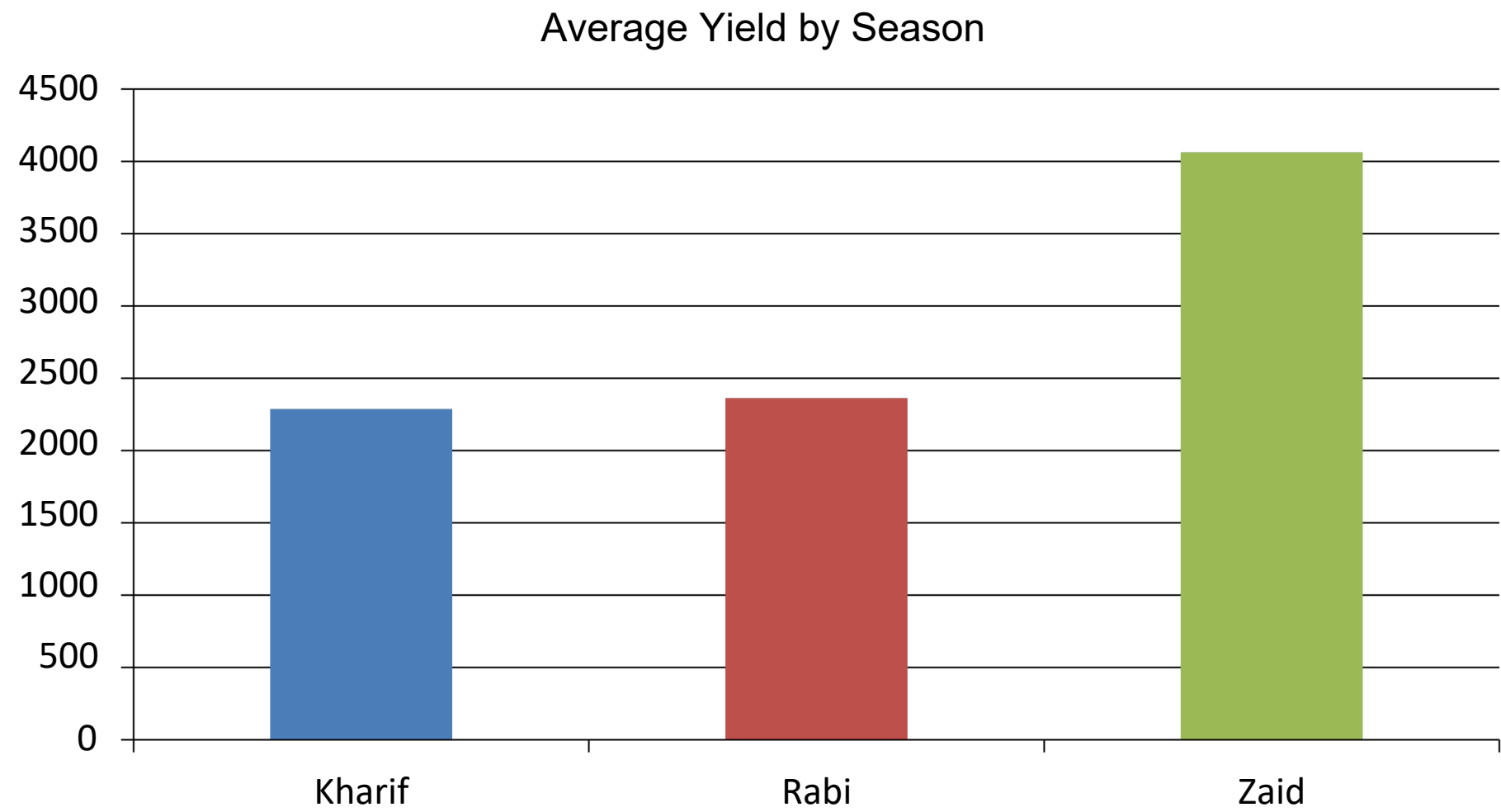
CSV –simple tabular data storage.

Optional extension: Streamlit or Power BI can be used to convert the analysis into an interactive dashboard.

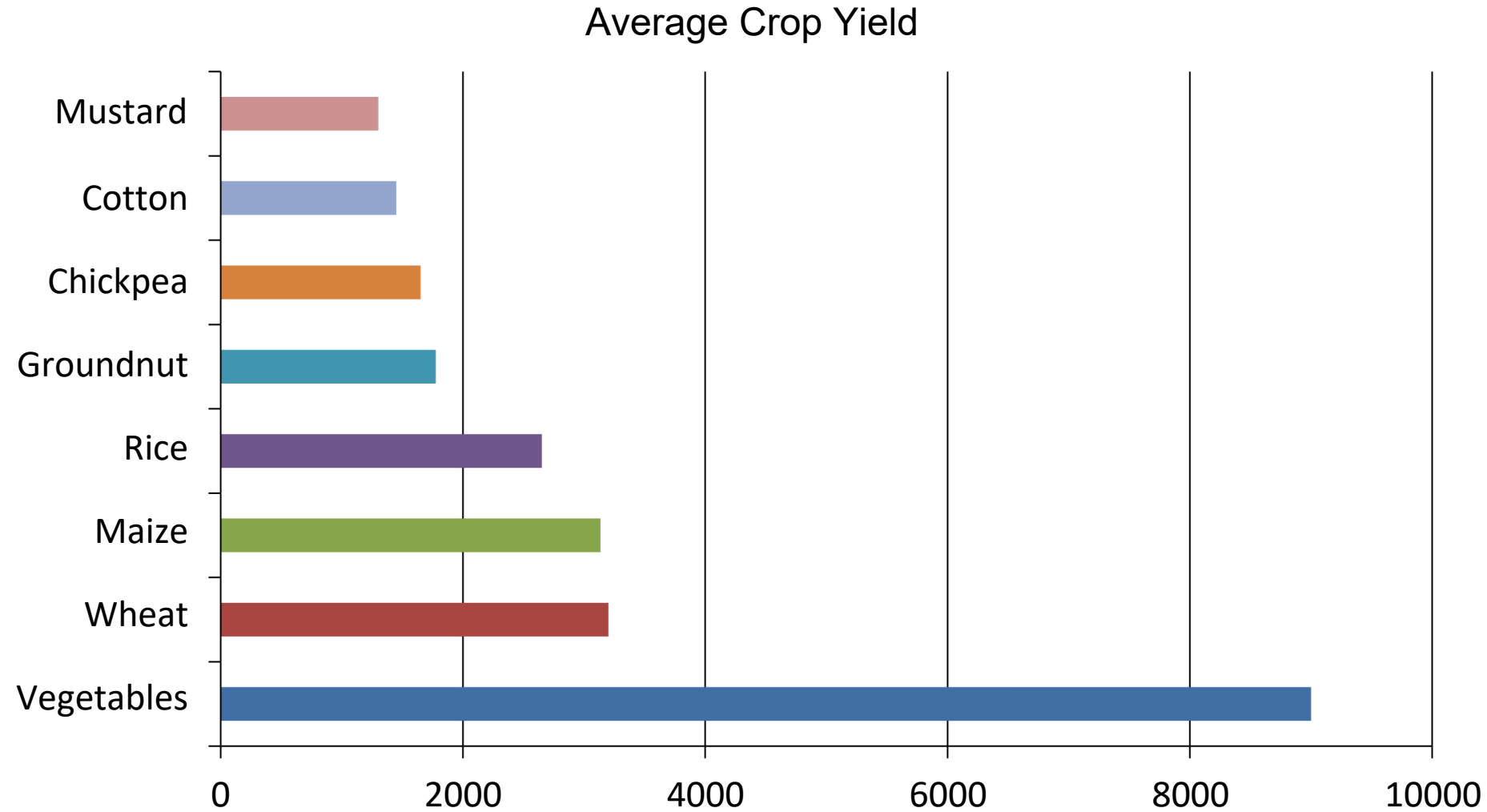
RESULTS –PRODUCTION BY SEASON



RESULTS –AVERAGE YIELD



RESULTS –CROP PERFORMANCE



RESULTS –WEATHER & YIELD

Rainfall and temperature are included as explanatory variables.

The scatter plot in the Python code can be used to inspect whether higher rainfall is associated with higher yield.

Because this is a sample educational dataset, the relationship should not be treated as a real-world causal conclusion.

Season	Avg. Rainfall (mm)	Avg. Temperature (°C)
Kharif	728	25.4
Rabi	378	19.8
Zaid	310	28.2

PROJECT WORKFLOW

- 
1. Collect / prepare agriculture data.
 2. Store the records in CSV format.
 3. Load the dataset using Pandas.
 4. Check columns and clean missing or inconsistent values.
 5. Group data by season and crop.
 6. Calculate production and average yield.
 7. Visualize results using bar charts and scatter plots.
 8. Interpret patterns and prepare recommendations.

FUTURE SCOPE



Connect the project to live or official agriculture datasets.

Add district-wise and state-wise comparisons.

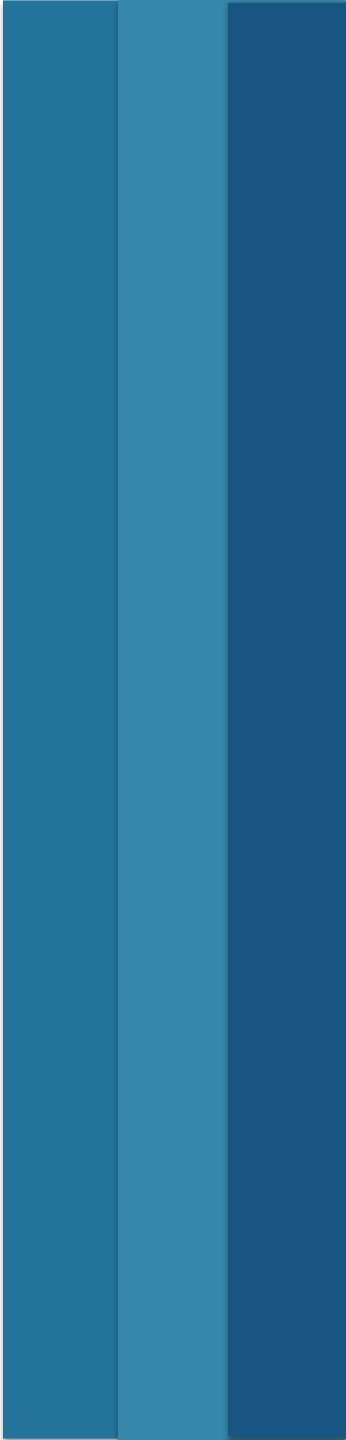
Include soil type, irrigation, fertilizer and market-price variables.

Build an interactive dashboard using Streamlit or Power BI.

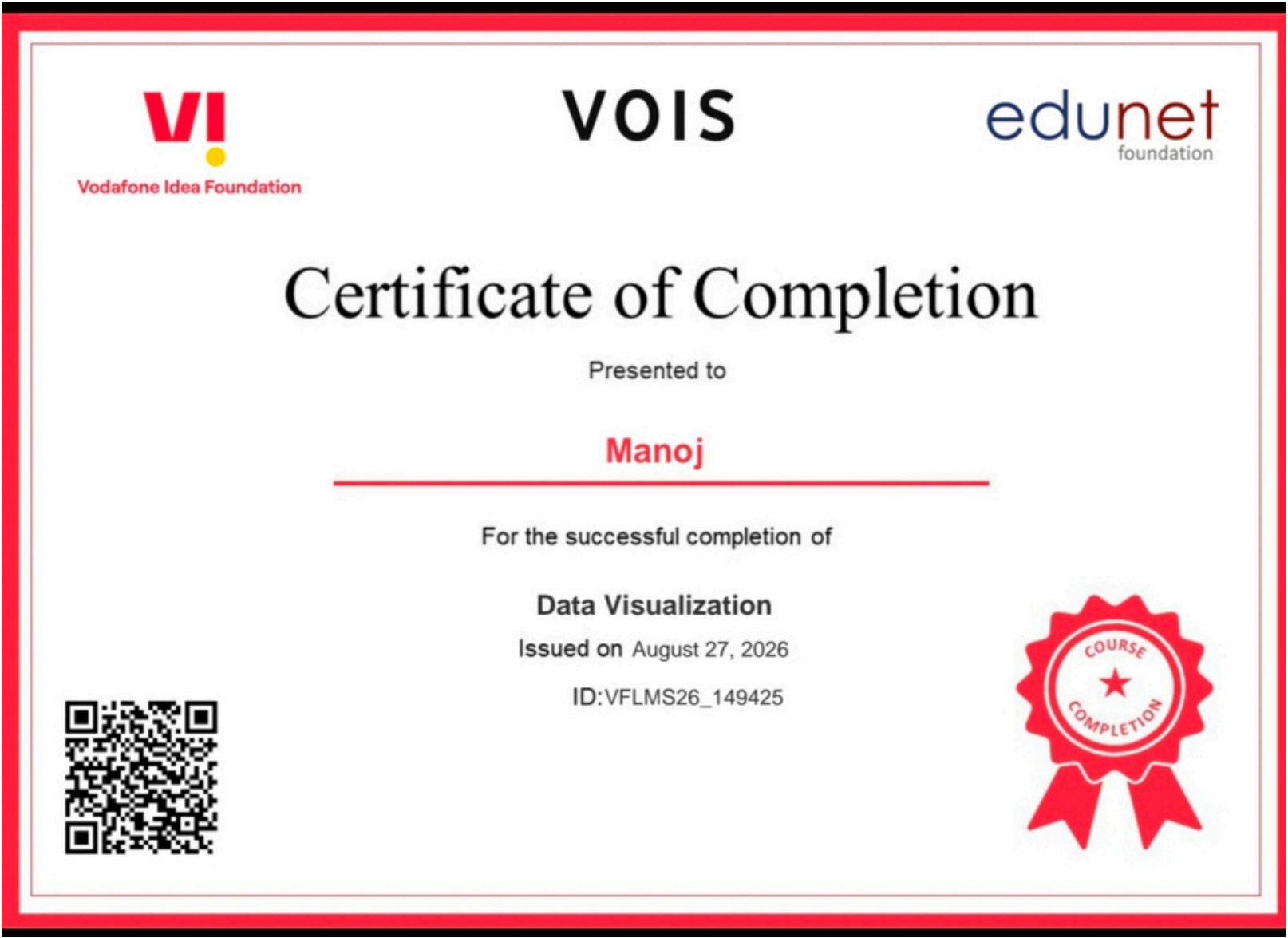
Add machine-learning models for yield prediction and crop recommendation.

Add yearly trend analysis and weather forecasts.

GITHUB LINK



VOIS Course completion certificates
Course name- Data Visualization



THANK YOU



Thank You
Questions & Discussion

